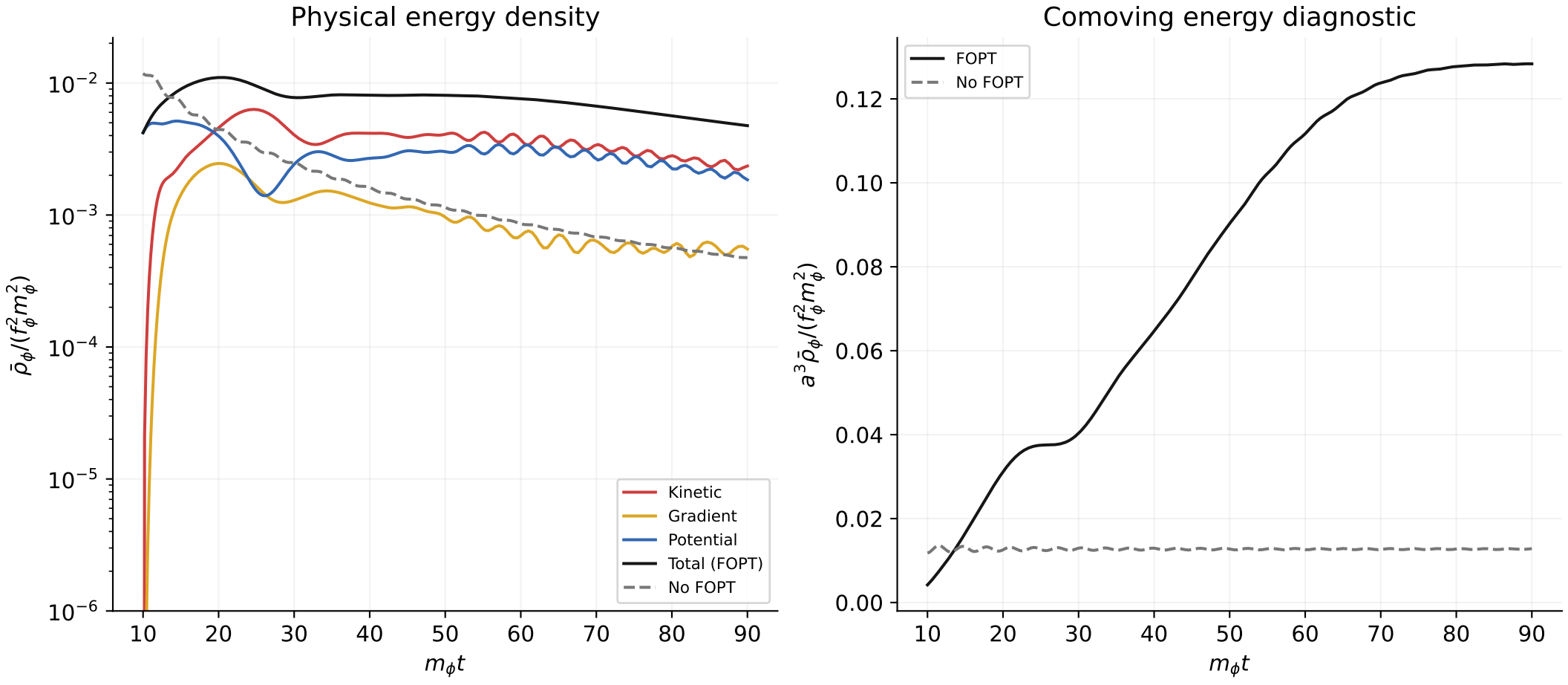
Physical energy density
$\bar{\rho}_\phi/(f_\phi^2 m_\phi^2)$
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10
20
30
40
50
60
70
80
90
$m_\phi t$
Kinetic
Gradient
Potential
Total (FOPT)
No FOPT
Comoving energy diagnostic
$a^3\bar{\rho}_\phi/(f_\phi^2 m_\phi^2)$
0.00
0.02
0.04
0.06
0.08
0.10
0.12
10
20
30
40
50
60
70
80
90
$m_\phi t$
FOPT
No FOPT